Supplementary Information for
“Assembly Does Not Identify Selection”

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S1 Assembly observables

Definition S1.1 (Assembly index and ensemble assembly)**.** Fix an assembly space with a finite nonempty basis. The *assembly index* a_i of object type i is the minimum number of nonbasic vertices in a rooted assembly subspace containing that type. Equivalently, it counts the joins in a shortest reusable construction pathway. Its *copy number* n_i is the number of observed occurrences of that type. For N observed types and N_T observed objects, define

$$A = \sum_{i=1}^{N} \mathrm{e}^{a_i} \left(\frac{n_i - 1}{N_\mathrm{T}} \right),$$

where e is Euler’s number [1–3].

Interpretation. Assembly index counts joins along one shortest permitted construction while allowing produced types to be reused. Copy number counts observed occurrences. Ensemble assembly combines these endpoint measurements. The reusable type pool used to calculate assembly index is distinct from the active physical occurrences used by a production law.

Definition S1.2 (Unary-chain assembly space)**.** Fix an integer $a \geq 2$. Let Γ_a have vertex set

$$V_a = \{1, \ldots, 2^a\}.$$

A vertex denotes a unary chain and its integer label denotes its length. For every $1 \leq r \leq s \leq 2^a$ with $r + s \leq 2^a$, include an edge from r to $r + s$ labelled s and an edge from s to $r + s$ labelled r. When $r = s$, one edge labelled r satisfies both roles.

Every edge strictly increases chain length, so Γ_a is acyclic. Its basis is $\{1\}$. Every vertex $m > 1$ has an incoming edge from 1 labelled $m - 1$, and every vertex is reachable from the basis. The paired edge labels satisfy the standard assembly-space axiom [3]. Hence Γ_a is a finite assembly space.

S2 Production laws

Definition S2.1 (Active inventory)**.** An *active inventory* is a finite multiset of physical object occurrences. A production join removes two active occurrences and inserts one occurrence of their joined type. The inserted occurrence is the *newest aggregate*. A production history ends when the inventory contains one occurrence. Only that terminal occurrence is observed.

Definition S2.2 (Preference ratio)**.** In the production process below, every unordered pair of distinct active occurrences is permitted. Fix a finite number $\lambda \geq 1$. The first pair is chosen uniformly. Suppose a later state has $k \geq 3$ active occurrences and newest aggregate x. Each of the $k-1$ pairs containing x receives weight λ. Each of the remaining $\binom{k-1}{2}$ pairs receives weight one. The conditional probability of a particular pair is therefore

$$\frac{\lambda}{\lambda(k-1)+\binom{k-1}{2}}$$

when it contains x, and

$$\frac{1}{\lambda(k-1)+\binom{k-1}{2}}$$

otherwise. When $k=2$, the sole pair has probability one. Define also the *newest-only law.* After the first join, its conditional probability for a particular pair is

$$\frac{1}{k-1}$$

when the pair contains x, and zero otherwise.

Interpretation. The preference ratio is the conditional weight of a pair containing the newest aggregate divided by the weight of another pair. When $\lambda = 1$, every pair has probability

$$\frac{1}{\binom{k}{2}},$$

so joining is equiprobable. When $\lambda > 1$, each pair containing the newest aggregate has greater probability than each other pair whenever both classes exist. Every pair retains positive probability under the finite family. The newest-only law reproduces the pair-selection pattern of the directed polymer example of Sharma et al. [1]. Their process retains input objects. Definition S2.2 governs consumptive physical occurrences.

S3 Non-identifiability

Definition S3.1 (Identifiability)**.** Let Y be an observation generated by a model indexed by a parameter θ. The parameter is *identifiable from* Y when

$$\mathcal{L}_\theta(Y) = \mathcal{L}_{\theta'}(Y) \quad \Longrightarrow \quad \theta = \theta',$$

where $\mathcal{L}_\theta(Y)$ denotes the probability law of Y under θ.

Interpretation. A parameter is identifiable when different parameter values cannot produce the same observation distribution.

Lemma S3.1 (Assembly index of the target)**.** *The assembly index of vertex* 2^a *in* Γ_a *is* a*.*

Proof. The rooted subspace on

$$\{1, 2, 4, \ldots, 2^a\}$$

constructs the target by repeated doubling and contains a nonbasic vertices. Hence the assembly index is at most a.

For the lower bound, take any rooted assembly subspace containing 2^a and topologically order its nonbasic vertices. After the first j nonbasic vertices in that order, every available chain has length at most 2^j. The claim holds for $j = 0$ because the only basic chain has length 1. If it holds after $j-1$ vertices, the next vertex joins two earlier chains, each of length at most 2^{j-1}, and therefore has length at most 2^j. A subspace containing length 2^a must consequently contain at least a nonbasic vertices. The assembly index is exactly a. □

Lemma S3.2 (Endpoint independence)**.** *Begin with 2^a active monomers in Γ_a. Under any possibly state-dependent or history-dependent law that selects one active pair at every nonterminal state, the process terminates after $2^a - 1$ joins at one chain of length 2^a.*

Proof. At every state, all active lengths are positive and sum to 2^a. Hence any active pair of lengths r and s satisfies $r + s \leq 2^a$ and is permitted by Definition S1.2. Each join preserves total length and reduces the occurrence count by one. Starting from 2^a occurrences, the process therefore reaches one occurrence after exactly $2^a - 1$ joins. Its length must equal the conserved total 2^a. □

Theorem S3.1 (Endpoint assembly observables do not identify preferential joining)**.** *For every pair of integers $a, n \geq 2$ and every finite $\lambda \geq 1$, the assembly space Γ_a and the corresponding production law in Definition S2.2 produce an observed ensemble with*

$$N = 1, \qquad a_1 = a, \qquad n_1 = N_{\mathrm{T}} = n, \qquad A = \mathrm{e}^a \frac{n-1}{n}$$

with probability one. The law is equiprobable when $\lambda = 1$ and preferential when $\lambda > 1$. The newest-only law produces the same ensemble with probability one.

Proof. By Lemma S3.1, the terminal type of length 2^a has assembly index a. By Lemma S3.2, one compartment terminates at that type on every history.

At $\lambda = 1$, the denominator in Definition S2.2 is

$$(k-1) + \binom{k-1}{2} = \binom{k}{2},$$

so every active pair has conditional probability $1/\binom{k}{2}$.

At every finite $\lambda > 1$, the state immediately after the first join has $2^a - 1 \geq 3$ occurrences. It contains the newest length-2 aggregate and at least two monomers. A pair containing the newest aggregate has conditional probability

$$\frac{\lambda}{\lambda(k-1) + \binom{k-1}{2}},$$

while a pair of monomers has conditional probability

$$\frac{1}{\lambda(k-1) + \binom{k-1}{2}}.$$

The former is greater. The production law is therefore preferential at a state reached with probability one. Under the newest-only law, some pair containing the newest aggregate is selected with probability one. Lemma S3.2 shows that every one of these laws has the same length-2^a endpoint.

Run n noninteracting copies of the compartment with the same inventory and stopping rule. Compartments do not exchange aggregates. Their terminal chains persist until observation, and intermediate occurrences have been consumed. The observed ensemble contains n copies of one type. Thus

$$N = 1, \qquad a_1 = a, \qquad n_1 = N_{\mathrm{T}} = n.$$

Substitution into the assembly equation gives

$$A = \mathrm{e}^a \frac{n-1}{n}$$

for every finite $\lambda \geq 1$ and for the newest-only law. $\square$

Let $y_{a,n}$ denote the terminal ensemble of n length-2^a chains, and let Y be the complete observed terminal ensemble. Theorem S3.1 gives

$$\mathcal{L}_\lambda(Y) = \delta_{y_{a,n}} \qquad \text{for every finite } \lambda \geq 1,$$

where $\delta_{y_{a,n}}$ is the point mass at $y_{a,n}$. The newest-only law induces the same point mass.

Corollary S3.1 (Production-law preference is not identified)**.** *Neither λ nor the proposition $\lambda > 1$ is identifiable from the complete terminal ensemble on any process class containing the finite-ratio laws in Theorem S3.1. The ensemble also does not distinguish the equiprobable law from the newest-only law. The same holds for every statistic of the endpoint pairs (a_i, n_i), including A.*

Proof. Fix a and n. The observation law is the same point mass for $\lambda = 1$, every finite $\lambda > 1$, and the newest-only law. The map from finite λ to the law of Y is therefore constant rather than one-to-one. Any statistic of Y also has the same law under these production laws. $\square$

Corollary S3.2 (No process-independent finite threshold)**.** *For fixed $n \geq 2$, equiprobable joining attains arbitrarily large values of A in this process class.*

Proof. Set $\lambda = 1$. Theorem S3.1 gives

$$A = \mathrm{e}^a \frac{n-1}{n},$$

which is unbounded as a increases. $\square$

S4 Scope

The theorem concerns inference from endpoint assembly observables over a process class containing the stated aggregation laws. It has six scope limits.

1. The active production inventory is consumptive. The product becomes available for later joins, while its inputs cease to exist as separate active occurrences. The illustrative forward exploration in Sharma et al. [1] retains its inputs and adds each product to the pool. The two processes use the same type-level joining relation and newest-product bias, but they are different occurrence laws.

2. The observation protocol records only terminal aggregates after a stated stopping rule. Intermediate histories or temporal observations can distinguish the production laws.

3. The monomer inventory is 2^a for a target with assembly index a. The unbounded sequence therefore does not keep its resource budget fixed.

4. The theorem does not refute reported molecular correlations or a classifier restricted to a specified chemical transition model. Such a model may exclude reconvergent consumptive processes.

5. The theorem establishes non-identifiability on a model class containing the paired laws. It does not claim that endpoint data can never alter posterior odds in a larger model. Any such update depends on the model and prior. The paired laws themselves have equal endpoint likelihood.

6. If every boundary condition, inventory constraint, or stopping condition is defined as selection, then the equiprobable process is selected under that definition. Selection then denotes constrained production generally. The common value of A still does not identify or quantify the added pair-choice preference.

The result establishes that transition probabilities, resource bounds, observation times, sampling rules, and prior assumptions are additional premises. They are not determined by assembly index, copy number, or ensemble assembly.

References

[1] Abhishek Sharma, Dániel Czégel, Michael Lachmann, Christopher P. Kempes, Sara I. Walker, and Leroy Cronin. Assembly theory explains and quantifies selection and evolution. *Nature*, 622(7982):321–328, 2023. doi: 10.1038/s41586-023-06600-9. URL `https://doi.org/10.1038/s41586-023-06600-9`.

[2] Christopher P. Kempes, Michael Lachmann, Andrew Iannaccone, G. Matthew Fricke, M. Redwan Chowdhury, Sara I. Walker, and Leroy Cronin. Assembly theory and its relationship with computational complexity. *npj Complexity*, 2:27, 2025. doi: 10.1038/s44260-025-00049-9. URL `https://doi.org/10.1038/s44260-025-00049-9`.

[3] Stuart M. Marshall, Douglas G. Moore, Alastair R. G. Murray, Sara I. Walker, and Leroy Cronin. Formalising the pathways to life using assembly spaces. *Entropy*, 24(7):884, 2022. doi: 10.3390/e24070884. URL `https://doi.org/10.3390/e24070884`.